

INTRODUCTION

THE PROBLEM

Electricity distribution systems generate large volumes of smart-meter data, making manual identification of abnormal consumption patterns and short-term demand behaviour difficult.

THE SOLUTION

MeterIQ analyses smart-meter consumption data to surface unusual patterns, potential electricity-theft risk signals, short-term load forecasts, and explainable insights via an interactive dashboard.

KEY OBJECTIVES

Identify abnormal consumption behaviour and potential risk signals

Forecast short-term electricity demand

Provide explainable decision-support insights

THE METHODOLOGY

1. Data Source

- Smart-meter consumption data
- Mathura & Bareilly distribution grids
- Timestamped electricity measurements

2. Preprocessing

- Missing-value handling
- Duplicate removal
- Data validation
- Timestamp alignment/resampling

3. Feature Engineering

- Lag features
- Rolling statistics
- Temporal features
- Consumption-based features

4. ML Analytics

- | | |
|--|---|
| Anomaly Detection | Load Forecasting |
| • Identify abnormal consumption patterns | • Short-term/24-hour demand prediction |
| • Generate anomaly score | • Forecast evaluation using error metrics |
| • Assign risk level | |

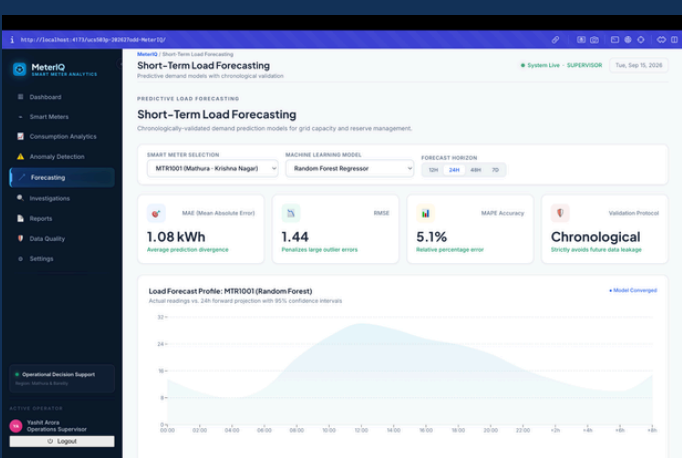
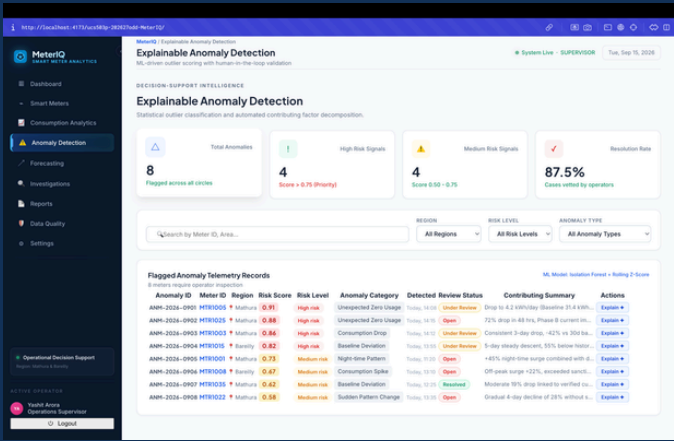
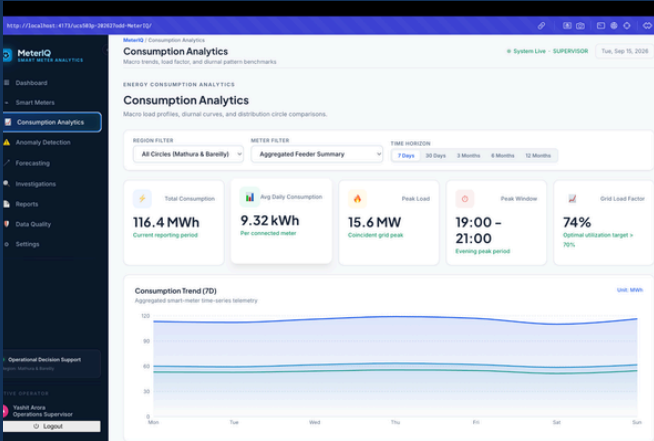
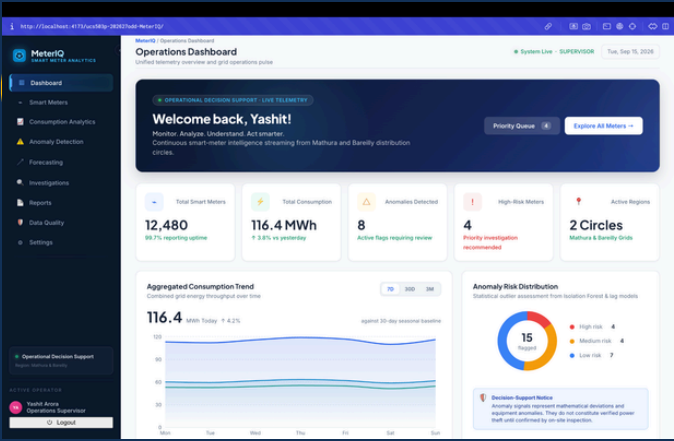
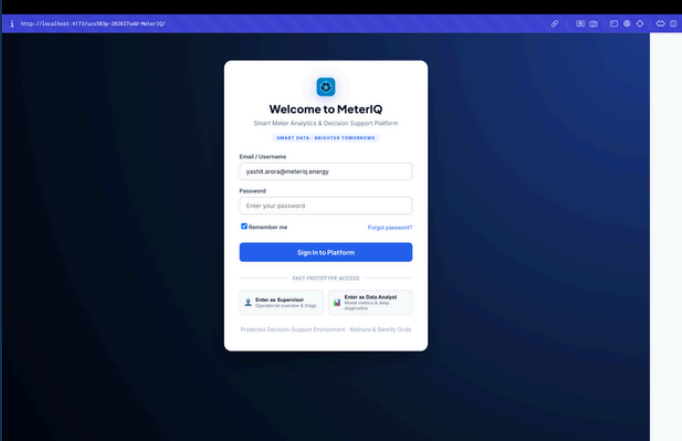
5. Explainability

- Identify contributing features
- Provide anomaly explanation
- Present supporting evidence/risk signals

6. Dashboard

- Consumption analytics
- Anomaly monitoring
- Forecast visualization
- Investigation support

Prototype



From Reactive Inspection → Data-Driven Decisions

◆

Anomaly Explanation: MTR1005

Record: ANM-2026-0901 · Mathura Circle

×

Calculated Risk Score

0.91

High risk

Anomaly Classification

Unexpected Zero Usage

Detection Timestamp

Today, 14:08

Why was this meter flagged?

The Isolation Forest ensemble and rolling statistical baseline identified high multi-dimensional variance in this meter's telemetries:

PRIMARY CONTRIBUTING FACTORS:

Drop to 4.2 kWh/day (Baseline 31.4 kWh), night zero-usage, 3.2σ peer divergence

RECOMMENDED HUMAN ACTION:

"Physical CT connection audit. Review recent smart-meter readings and compare against feeder peer meters."

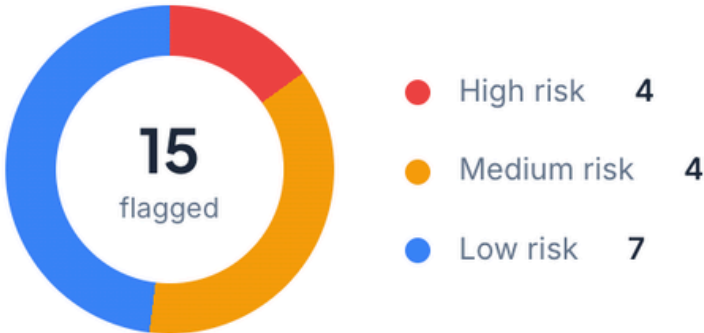
Final determination requires human operational review.

Full Meter Diagnostics

Dispatch to Investigation Queue →

Anomaly Risk Distribution

Statistical outlier assessment from Isolation Forest & lag models



Decision-Support Notice

Anomaly signals represent mathematical deviations and equipment anomalies. They do not constitute verified power theft until confirmed by on-site inspection.